# Proprioceptive Surprise-Based Episodic Memory for Test-Time Adaptation in Long-Horizon Manipulation

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**Abstract:**
A significant open challenge in robot learning is the development of policies that can adapt through interaction with unpredictable real-world environments. We explore proprioceptive prediction error as a signal for test-time memory writes. Inspired by evidence that dopaminergic prediction error signals modulate the strength of episodic memory formation, we use this signal to guide test-time adaptation of a low-rank episodic memory tensor within a pretrained vision-language-action model, drawing loosely on the fast episodic system of complementary learning systems theory. We introduce SOAP (Surprise-based Online Adaptation using Proprioception), a family of write rules that vary which aspects of the surprise gradient they retain, and evaluate it on LIBERO-Long across three repeats of 500 rollouts. SOAP-Direction, which retains only the gradient's direction by normalizing it to a fixed norm, achieves a 95.4% success rate against a 93.2% baseline (+2.2 percentage points, 95% CI 0.7 to 3.7), and outperforms random directions at equal gradient norm (93.9%). SOAP-Full, which retains both direction and magnitude, reaches 94.2%, within noise of the baseline. SOAP-Direction was identified in an exploratory analysis and requires held-out replication, but suggests that the gradient's direction may be more informative than its magnitude. All results are obtained under 4-bit quantization on a consumer-grade NVIDIA RTX 4070 Super, indicating feasibility in compute-constrained settings where high control frequency is required.

**Keywords:** Proprioceptive Surprise, Test-Time Adaptation, Episodic Memory, Complementary Learning Systems

## 1 Introduction

Vision-language-action (VLA) models have demonstrated remarkable success in generalizing to unseen task environments, leveraging large-scale pretraining to ground natural language instructions to low-level visuomotor behavior. Despite these advances, VLAs continue to struggle with long-horizon task execution, which is a limitation rooted not purely in linguistic or perceptual capacity, but in the absence of persistent memory across task sequences. On benchmarks such as LIBERO-Long, which demands long-horizon sequential manipulation, this deficiency manifests itself concretely: a robot that has placed a bowl in a drawer has no mechanism for remembering having done so, and one that drops a scooper has no mechanism to weight that failure more heavily than a successful grasp. While the former remains a broader open problem in object state tracking, the latter reflects a failure of memory selectivity, which we aim to address in this work.

Standard approaches to memory in learned robot policies, such as recurrent architectures [1, 2], explicit memory modules, and extended context methods [3], while effective in specific settings, fail to provide a principled mechanism for distinguishing between what information is worth remembering and what is not. In contrast, neuroscience provides a biologically grounded framework for selective

memory encoding: the complementary learning systems (CLS) theory posits that the human brain relies on both a fast-learning episodic hippocampal memory system and a slow-learning neocortical system for general knowledge [4]. Specifically, dopaminergic prediction error signals are thought to modulate the strength of hippocampal encoding [5, 6]. We propose **SOAP**: Surprise-based Online Adaptation using Proprioception, a family of low-rank memory write rules driven by proprioceptive prediction error, enabling test-time adaptation without modifying the pretrained weights or introducing retrieval overhead. Rather than assuming which aspect of surprise matters, we compare variants that retain or remove the surprise gradient's direction and magnitude.

## 2 Method

We formulate SOAP as a three-timescale decomposition of effective weights. As illustrated in Figure 1, we augment the pretrained OpenVLA-OFT backbone [7] with a lightweight episodic memory delta updated at test time. Specifically, SOAP variants derive writes to this delta from proprioceptive prediction error, a surprise signal inspired by dopaminergic prediction error signaling.

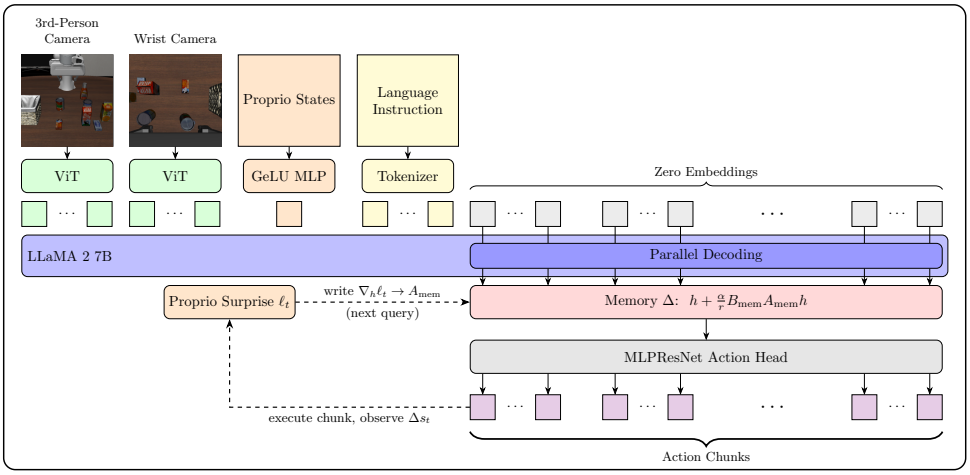



Figure 1: Overview of the SOAP architecture. The pretrained OpenVLA-OFT backbone produces action hidden states, which are adapted at test time by the episodic memory delta $B_{\text{mem}}A_{\text{mem}}$ before the action head. Dashed arrows show the write path: after each action chunk executes, the proprioceptive surprise gradient $\nabla_h \ell_t$ is written into $A_{\text{mem}}$ at the next query, after that query's actions are predicted, and affects actions from the following query onward.

### 2.1 Three-Timescale Weight Decomposition

The timescale-decomposition of SOAP is represented as:

$$W_{\text{eff}} = \underbrace{W_0}_{\text{pretrained}} + \underbrace{BA}_{\text{task LoRA}} + \underbrace{\frac{\alpha_{\text{mem}}}{r} B_{\text{mem}} A_{\text{mem}}}_{\text{episodic delta}},$$

where $W_0$ is the frozen pretrained backbone, $BA$ is the frozen task-specific LoRA (fixed after fine-tuning), and $B_{\text{mem}}A_{\text{mem}}$ is the episodic memory delta updated only at test time. This decomposition is conceptual: in practice, the episodic delta is applied to the action hidden states before the action head (Figure 1), not to the backbone weights. The pretrained backbone already spans the relevant semantic directions for manipulation, whereas the fine-tuning LoRA matrices identify the subspace relevant to a specific task. We constrain $B_{\text{mem}}$ and $A_{\text{mem}}$ to rank $r$ so that online updates remain small and cheap. For the sake of experimentation, we set $B_{\text{mem}}$ to a frozen random projection rather than deriving it from the task LoRA; a learned subspace is a prudent area for future exploration.

### 2.2 Instantiating Proprioceptive Surprise

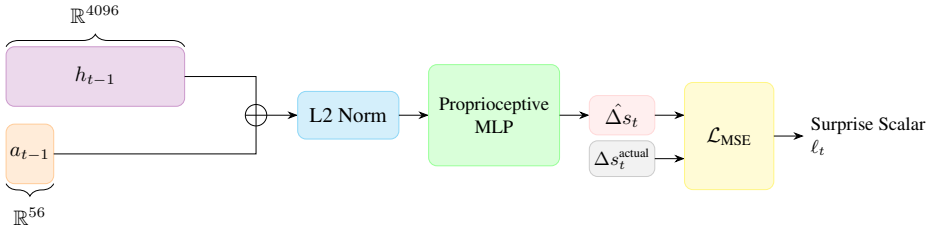



Figure 2: Proprioceptive prediction and surprise value determination.

To properly update the episodic memory delta at test time, SOAP requires a scalar signal indicating the degree to which a robot's state transition was surprising. Figure 2 depicts the offline-trained proprioceptive prediction head used to forecast robot state transitions at each timestep, where prediction error between predicted and actual state transition is used as a proxy for proprioceptive surprise. The proprioceptive prediction block takes $h_{t-1}$ and $a_{t-1}$ as input. $h_{t-1}$ is Llama 2's final layer mean-pooled hidden state from within OpenVLA-OFT, whereas $a_{t-1}$ is the action chunk predicted at the prior timestep. Both are concatenated prior to being normalized and passed through the MLP, conditioning surprise on both the model's internal representation and its predicted action, since the same perceived state may yield different transitions depending on what the robot does. Intuitively, consider a grasp attempt in which the end-effector makes unexpected contact with an obstacle. The resulting state transition deviates significantly from the proprioceptive head's predicted state transition, yielding a high surprise signal and, when gradient magnitude is retained, a large memory write. Conversely, a successful transition proceeds as predicted, producing a low surprise signal and a negligible write. Formally:

$$\ell_t = \frac{1}{d_s} \left\| \text{ProprioHead}\left( \frac{[h_{t-1} \| a_{t-1}]}{\| [h_{t-1} \| a_{t-1}] \|_2} \right) - \frac{\Delta s_t - \mu}{\sigma} \right\|_2^2$$

### 2.3 Surprise-Based Memory Writes

The memory delta term is updated at each timestep via a Titans-style write [8]. The momentum accumulator is updated as $\mathbf{S}_t = \eta_s \mathbf{S}_{t-1} - \theta_s \nabla_h \ell_t$, where $\nabla_h \ell_t$ is the gradient of the proprioceptive surprise loss with respect to the adapted hidden state. $\mathbf{S}_t$ accumulates surprise forward via $\eta_s$, so a sequence of surprising steps compounds rather than each one being treated independently. $A_{\text{mem}}$ is then written to as $A_{\text{mem}} \leftarrow \lambda A_{\text{mem}} + \eta \left( B_{\text{mem}}^\top \mathbf{S}_t \right) \otimes h_{t-1}$, where $\lambda$ is an exponential decay factor and the outer product produces a rank-1 update of shape $\mathbb{R}^{r \times d}$. To prevent memory from leaking into subsequent action episodes, $A_{\text{mem}}$ is zeroed at the start of each trial. When gradient magnitude is retained, selectivity is soft rather than binary: because $\nabla_h \ell_t$ is the proprioceptive prediction residual back-propagated through the prediction head, its magnitude grows with the prediction error, so surprising transitions produce large writes to $A_{\text{mem}}$ while well-predicted transitions contribute little. An optional hard threshold $\tau$ can additionally skip writes for which $\ell_t < \tau$; we set $\tau = 0$ in all experiments.

## 3 Experimental Results

### 3.1 Experimental Setup

We evaluate the SOAP suite on LIBERO-Long, a simulation benchmark of ten long-horizon manipulation tasks, each requiring sequential execution of multiple subtasks within a single rollout. We compare a vanilla OpenVLA-OFT *Baseline* with three SOAP variants and a control: *SOAP-Full* writes $\nabla_h \ell_t$ unchanged; *SOAP-Direction* rescales it to a fixed norm of 0.215 (its mean over a 20-episode pilot); *SOAP-Magnitude* keeps its norm but uses a random Gaussian direction; and

*Random* uses a random direction at the fixed norm, with no per-step surprise information. Each runs 50 rollouts per task across three repeats, which reuse LIBERO's fixed initial states, so we average outcomes per state and test paired differences over the 500 states with two-sided sign-flip tests.

### 3.2 Performance Evaluation

| | Task | Baseline | SOAP-Full | SOAP-Direction | SOAP-Magnitude | Random |
|---|---|---|---|---|---|---|
| 0 | Put alphabet soup and tomato sauce in basket | 42.0 | 43.0 | **44.0** | 42.0 | 41.3 |
| 1 | Put cream cheese box and butter in basket | 49.0 | 49.0 | **50.0** | 49.7 | 49.3 |
| 2 | Turn on stove and put moka pot on it | **49.7** | **49.7** | 49.3 | **49.7** | **49.7** |
| 3 | Put black bowl in bottom cabinet drawer and close | 49.0 | 49.0 | **49.7** | 48.0 | 48.3 |
| 4 | Put white mug on left plate, yellow mug on right | **50.0** | **50.0** | **50.0** | **50.0** | **50.0** |
| 5 | Pick up book and place in back caddy compartment | 49.7 | 49.7 | **50.0** | 49.7 | 49.7 |
| 6 | Put white mug on plate, pudding to right of plate | 39.0 | 42.0 | **44.0** | 42.3 | 43.3 |
| 7 | Put alphabet soup and cream cheese in basket | 48.0 | 48.0 | 47.0 | 48.3 | **48.7** |
| 8 | Put both moka pots on stove | 43.3 | 44.3 | **47.0** | 44.7 | 44.3 |
| 9 | Put yellow mug in microwave and close it | **46.3** | **46.3** | 46.0 | 45.0 | 44.7 |
| | **Mean total (/500)** | 466.0 | 471.0 | **477.0** | 469.3 | 469.3 |

Table 1: Mean per-task successes (out of 50, three repeats) for each variant. Total ranges, in column order: 461–469, 470–472, 475–478, 467–471, 468–471.

As shown in Table 1, SOAP-Direction improves on the baseline by 2.2 percentage points (95.4% vs. 93.2%; bootstrap 95% CI 0.7 to 3.7, $p = 0.005$) and outperforms SOAP-Magnitude and Random (93.9% each; $p = 0.011$ and $p = 0.018$). SOAP-Full (94.2%, $p = 0.18$), SOAP-Magnitude, and Random ($p \approx 0.4$) do not separate from the baseline, and SOAP-Direction does not separate from SOAP-Full ($p = 0.082$), so surprise-scaled strength shows no measurable benefit in this implementation. The directional advantage depends on magnitude: SOAP-Full does not outperform SOAP-Magnitude ($p = 0.63$). The gradient may therefore provide a useful write direction at fixed magnitude; we do not claim that direction matters in general, and we treat the hippocampal account as design motivation, not an explanation of these results. An earlier single-repeat evaluation reported 95.2% for SOAP-Full against 92.0% for the baseline and 16.4% for unnormalized Gaussian writes of norm $\approx 64$, suggesting that write scale is critical.

**Limitations.** SOAP-Direction was singled out and the state-level test chosen after inspecting data, so this analysis is exploratory and needs held-out replication. Excluding task 6 raises its $p$-value against the baseline to 0.07. Claims are limited to LIBERO's fixed initial states. Fixture poses on tasks 2, 3, 5, 8, and 9 re-randomize at each reset, so those pairs are not scene-identical, and first-repeat variants ran in separate sessions. All runs share one frozen random $B_{\text{mem}}$, preserving about $r/d \approx 0.2\%$ of the gradient's energy, and we did not test whether writes reduce error. Further experimentation is also needed against state-of-the-art test-time adaptation mechanisms, on less saturated benchmarks than LIBERO-Long, where most tasks are near ceiling, and across the sim-to-real gap.

## 4 Conclusion

We presented SOAP, a family of lightweight test-time write rules for pretrained VLAs, comparing variants that retain the surprise gradient's direction, magnitude, or both. SOAP-Direction outperformed the baseline, SOAP-Magnitude, and Random on the standard initial states, while the surprise scaling in SOAP-Full showed no measurable benefit here. Future work should replicate these experiments on held-out states, test learned memory bases and surprise magnitudes beyond those used here, and explore consolidating the episodic delta into slow-learning components, akin to CLS-based consolidation.

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## A Implementation Details

$B_{\text{mem}} \in \mathbb{R}^{d\times r}$ is initialized as a frozen, $\ell_2$-normalized random Gaussian projection; $A_{\text{mem}} \in \mathbb{R}^{r\times d}$ is initialized to zero and reset at the start of every trial. Table 2 lists the memory rank, decay, and surprise-gating parameters used across all LIBERO-Long evaluations.

| Symbol | Description | Value |
|---|---|---|
| $r$ | Memory delta rank ($B_{\text{mem}}A_{\text{mem}}$) | 8 |
| $\alpha_{\text{mem}}$ | LoRA scaling, applied as $\alpha_{\text{mem}}/r$ | 8 |
| $\eta$ | Hebbian write learning rate ($A_{\text{mem}}$ update) | $1 \times 10^{-3}$ |
| $\lambda$ | Exponential forgetting decay on $A_{\text{mem}}$ per write | 0.99 |
| $\eta_s$ | Titans past-surprise momentum | 0.9 |
| $\theta_s$ | Titans momentary-surprise gradient scale | 1.0 |
| $\tau$ | Optional hard threshold on $\ell_t$ for $A_{\text{mem}}$ writes | 0.0 |

Table 2: Memory delta and surprise-gating hyperparameters used in all LIBERO-Long evaluations.

Writes are applied whenever $\ell_t \geq \tau$, so $\tau = 0.0$ applies a write at every timestep; in variants that retain gradient magnitude, selectivity then comes entirely from the magnitude of $\nabla_h \ell_t$ (Section 2.3).